

PROJECT REPORT

1. Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are changing how we approach everyday problems. One common issue students face is managing their study time properly. It's often confusing to decide which subject to focus on first, especially when deadlines are close and some subjects feel harder than others.

This project, AI-Based Smart Study Planner, tries to solve that problem by providing a smarter way to plan studies. Instead of manually guessing what to study, the system uses machine learning along with simple decision-making techniques to create a study plan based on difficulty and deadlines.

2. Problem Statement

Students commonly deal with a few major issues:

Not knowing which subject to prioritize

Poor time management, which leads to stress (especially before exams)

No proper or structured study plan

Difficulty balancing multiple subjects at the same time

Most students still rely on manual planning, which honestly isn't very efficient. So, there's a need for a system that can automatically suggest what to study and when.

3. Objective

The main goals of this project are:

To build an AI-based system that helps students organize their study schedule

To use machine learning for predicting subject priority

To create a simple decision-making system based on inputs

To design an easy-to-use interface

To show the study plan visually so it's easier to understand

4. Methodology

The project follows a step-by-step approach:

1. Data Collection

A dataset was created manually with values like subject difficulty, days left, and priority.

2. Data Processing

The dataset is loaded and divided into input features (difficulty, days left) and output (priority).

3. Model Training

A machine learning model is trained using this dataset.

4. User Input

Users enter subject name, difficulty level, and deadline.

5. Prediction

The model predicts how important (priority) each subject is.

6. Scheduling

Subjects are arranged in order of priority.

7. Visualization

The final plan is displayed with graphs for better clarity.

5. Algorithms Used

1. Linear Regression (ML Model)

This is used to predict the priority score.

Inputs:

Difficulty

Days left

Output:

Priority score

Basically, the model learns patterns from the data and tries to predict how urgent a subject is.

2. Heuristic-Based Sorting

After getting the priority scores, subjects are sorted so that higher priority ones come first.

This is a simple but effective approach—it's not too complex, but it works well for decision-making.

6. Implementation

The project is divided into three main parts:

1. Model Module (model.py)

Trains the machine learning model

Handles prediction logic

2. Planner Module (planner.py)

Uses the trained model

Generates the sorted study plan

3. User Interface (app.py)

Built using Streamlit

Takes user input

Displays results and graphs

Technologies Used:

Python

Pandas

Scikit-learn

Streamlit

Matplotlib

7. Results

The system was able to:

Take multiple subjects as input

Predict priorities for each subject

Generate a study plan automatically

Display results in sorted order

Show graphs for better understanding

Observation:

Subjects with higher difficulty and fewer remaining days were given higher priority, which makes sense practically.

8. Challenges

While working on the project, a few issues came up:

Import errors because of file/module structure

Problems with file paths while loading data

Handling empty or incorrect user inputs

Streamlit sometimes showing a blank screen (took time to debug)

Managing Python dependencies

These were fixed step-by-step by debugging and simplifying the project setup.

9. Conclusion

The AI-Based Smart Study Planner shows how AI and ML can actually be used in real-life situations like study planning.

It helps by providing:

Smarter decision-making

Better time management

A simple and interactive interface

Overall, this project connects well with core AI concepts like:

Intelligent agents

Machine learning models

Problem-solving techniques